

The empirical recurrence for OEIS A236038, recovered from its tail

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Abstract

OEIS A236038 records “Empirical recurrence of order 87” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 2404$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the whole sequence, so the recurrence holds from $n = 88$ — every index at which it can be written down. Its last coefficient is $c_{87} = 18479922936589320192 \neq 0$, so no further tail satisfies anything shorter; any order-87 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A236038 is “Number of $(n+1) \times (2+1)$ 0..3 arrays colored with the difference of the upper median and the lower median in each 2×2 subblock.”. It has offset 1 and begins

2404, 42264, 722548, 12435076, 215833320, 3742165712, 64753334132, 1120625605524, ...

The entry states:

Empirical recurrence of order 87 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:01:53, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 2404$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 2404$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Here the stated order is already the minimal order of the sequence itself, so no tail has to be taken.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 4808$ exact terms from $n = 1$ on, it returns order 87 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{87} c_i a(n-i),$$

c_1	0	c_{23}	−3097771895123106	c_{45}	−2788137278300769242209	c_{67}	3875143767
c_2	53	c_{24}	830251510641332	c_{46}	−3559027745600157225272	c_{68}	−4882108444
c_3	1645	c_{25}	−271920643917514	c_{47}	−3678670545166959751338	c_{69}	6211157790
c_4	27365	c_{26}	−20660894473879499	c_{48}	−467330513392032546854	c_{70}	21104060733
c_5	217281	c_{27}	63051821220649699	c_{49}	6493245860511458521520	c_{71}	11590755891
c_6	1408594	c_{28}	42581448218920152	c_{50}	12406128331826343152672	c_{72}	−1184140851
c_7	5668575	c_{29}	383494680519606	c_{51}	13983462005154521159358	c_{73}	5395033199
c_8	−7620655	c_{30}	672273890924558188	c_{52}	14178878884516038663245	c_{74}	8769921687
c_9	−178960497	c_{31}	−1408952247439254890	c_{53}	5405589729612279342413	c_{75}	−4450745971
c_{10}	−1014494025	c_{32}	−2617549307071833470	c_{54}	−18966511475943687062667	c_{76}	−8535838077
c_{11}	−3158130891	c_{33}	4491014432782304390	c_{55}	−40927884567690898751018	c_{77}	−672507798
c_{12}	6082229839	c_{34}	−1843500329273537328	c_{56}	−45520666036161497901732	c_{78}	1277308584
c_{13}	81763655461	c_{35}	529122797779970103	c_{57}	−28416623227547270130004	c_{79}	−413453177
c_{14}	234678684820	c_{36}	16650734856982168240	c_{58}	671274706632863592818	c_{80}	−564749953
c_{15}	107575758736	c_{37}	−41121100364726375538	c_{59}	38460866065869884594140	c_{81}	−391838387
c_{16}	−2885745294108	c_{38}	−57959347918918852694	c_{60}	57689826208676608314124	c_{82}	−314794342
c_{17}	−12290539561631	c_{39}	4746425238557122923	c_{61}	42656177334011934616976	c_{83}	−129619125
c_{18}	1953607919390	c_{40}	134239894344885451348	c_{62}	16623849483350828985824	c_{84}	398614693
c_{19}	75390873759907	c_{41}	419862409632117740117	c_{63}	−50170850269794875885104	c_{85}	563222064
c_{20}	99364396217348	c_{42}	601159660239359763354	c_{64}	−127508129483551314309376	c_{86}	190175907
c_{21}	224314966815752	c_{43}	377027741629571614430	c_{65}	−101454839585526171525312	c_{87}	184799229
c_{22}	−294447317247309	c_{44}	−1048954400010139582825	c_{66}	28438375997175304010496		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 88$, and it is the recurrence OEIS A236038 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 1$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{87} - c_1x^{87-1} - \dots - c_{87}$. Its constant term is $-c_{87} = -18479922936589320192$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and

only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 87 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 87, so $\deg q = 87$, and it is monic. It holds from some index t on. From $\max(t, 1)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 87, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 87, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 18479922936589320192.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-87 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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